

A CNN Centric Approach to Acute Lymphoblastic Leukemia Detection

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Abstract—Leukaemia is a serious malignancy that impacts both adults and children, presenting as either a chronic or acute condition. Acute Lymphoblastic Leukemia (ALL) stands as the most prevalent type of blood disease and remains a leading cause of death. This aggressive cancer invades the blood, spreading via surrounding bodily systems and organs. The conventional method for classifying cancer cells and non-cancer cells involves manual diagnosis, where specialists inspect cell images under a microscope and annotate the labels. However, this manual microscopic analysis is not only tedious but also susceptible to potential false diagnoses. In response to these challenges, this project seeks to leverage a pre-trained model for automated detection and classification of leukemia subtypes, including benign, early, pre, and pro stages. By utilizing a pretrained model, the proposed system aims to enhance its ability to accurately discern and classify leukemia subtypes. By leveraging pre-trained models, the proposed approach aims to improve classification accuracy, efficiency, and reduce the risk of misdiagnosis compared to manual analysis. This research contributes to the advancement of automated leukemia diagnosis, paving the way for potential clinical applications.

Index Terms—Leukemia, acute lymphoblastic leukemia (ALL), malignant illness, automated detection, microscopic analysis, diagnostic process, medical imaging, machine learning.

I. INTRODUCTION

The life-giving fluid that flows through our veins, blood, is made up of a complex ecosystem of cellular elements that are essential to many body processes. Among these elements, the delicate symphony of health is orchestrated by the equilibrium of red blood cells, white blood cells, and plasma. But when diseases like leukaemia upset this equilibrium, the results may be disastrous.

Leukemia, a formidable adversary, strikes at the very heart of the body's blood-producing machinery. Characterized by the unchecked proliferation of malignant white blood cells [14], it not only undermines the body's ability to fight infections but also infiltrates vital organs, posing a grave threat to life itself.

There are two primary forms of leukaemia: acute and chronic [4], [5], [7], [8]

ALL is a blood cell malignancy that proliferates swiftly. It starts in the bone marrow and eventually causes an unnatural increase of white blood cells (WBCs) [5], [7]–[10]. Targeting lymphocytes, key agents of the immune system, ALL can swiftly overwhelm the body if left unchecked, necessitating timely and precise diagnosis for successful treatment. Traditional methods of leukemia diagnosis, reliant on manual examination of blood and bone marrow samples by experienced hematopathologists, suffer from inherent limitations. Time-consuming, subjective, and prone to human error, these methods fall short in the face of the urgent need for swift and accurate diagnoses. In order to get around the aforementioned restrictions, research has shown a variety of computer-aided methods for ALL diagnosis that use microscopic blood image analysis to detect leukaemia. There have been several developments in medical research and diagnostic technologies in the last few years [5]–[12]. These techniques were discovered to be more effective, quick, accurate and more affordable than manual techniques. [1] proposes a segmentation algorithm that utilizes color and texture information to extract leukocytes, their cytoplasm and nucleus from pictures of bone marrow with heterogeneous staining. It suggests using conical shapes to separate overlapped blood elements. After segmentation, feature extraction is performed on the segmented regions. These extracted attributes are then used to classify the cells into leukemia subtypes. The study reports an average accuracy of 95% in the segmentation process and an overall accuracy of 92% in the acute leukaemia supervised categorization types, with 84% accuracy in lymphoblastic subtypes and 92% accuracy in myeloblastic subtypes [2]. The proposed methodology by [3] comprises two main components: automatic nucleus segmentation and CNN-based classification. The nucleus segmentation process is designed to accurately identify and isolate the nuclei of white blood cells, while the CNN-based classi-

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fication model is utilized to categorize the segmented cells into different classes. The research presented by Das PK in [4] focuses on developing an automated process for the detection as well as classification of acute lymphocytic leukemia (ALL). The study involves the extraction of lymphocytes using k-means clustering based on colour, then the segmented pictures' form, texture, and colour attributes are extracted. The techniques known as gray-level runlength matrix (GLRLM) and gray-level co-occurrence matrix (GLCM) are utilized for nucleus feature extraction, with PCA, or principal component analysis, is used to reduce dimensions. Mohammed and Abdulla [13], on the other hand, have shown an effective SVM-based ALL recognition system, extracting DCT-based characteristics in addition to spatial, geometric, and morphological information. A Graylevel co-occurrence matrix is applied in [10] to achieve feature extraction. Then, probabilistic PCA is used to reduce features. Using Random Forest, more effective ALL categorization is finally accomplished. After segmenting WBC cells and converting pictures to HIS colour space, Sampathila [15] trained and evaluated the ALLNET model, achieving a 95.54% accuracy and a 95.91% sensitivity. To train the ALL dataset, Sorayya in [16] adjusted both the weights and parameters of the VGG16 [23] and ResNet50 [22] models. They also suggested a convolutional network and six machine learning methods. Comprising a classification layer and ten convolutional layers. The accuracy of the VGG16 network was 84.62%, compared to 82.1% for the convolutional network. According to RF, the machine-learning algorithm's greatest accuracy was 81.72%. A successful transfer learning model has been suggested by Das [8] by marginally altering ResNet50 [22]. Subsequently, the model demonstrates superior training performance when used to extract important characteristics. Next, machine learning techniques are applied to obtain effective leukaemia detection. Leukocytes have been extracted from the background using the triangle approach for thresholding by Mishra [17] in 2019. A ShuffleNet [24] based ALL detection model was proposed by Das and Meher [18] in 2021. It retains the benefits of group convolution, depthwise separable convolution, and channel shuffling while producing promising results.

II. RESEARCH METHOD

In the field of medical imaging, transfer learning is becoming more popular due to its higher performance [8], [19]–[21]. Because a pre-trained network is used in transfer learning, and applies its expertise or weights to the tasks in the target domain. It has the benefit of not requiring a huge dataset for training. Therefore, all that is needed in the target area is for these pre-trained networks to be fine-tuned. As a result, it produces exceptional results even for small datasets [8], [19]–[21]. We employed the Resnet 50 and Inception models in this study. The ResNet model, developed by Microsoft Research, introduced skip connections to enable training of deeper neural networks without vanishing gradients [22]. Inception models, including InceptionV3 [25] by Google emphasizes multi-scale convolutions and efficient dimensionality reduction

for improved performance on assignments such as image classification. Preprocessing, both models' testing and training, as well as an examination and comparison of the results is one of the main stages of this research.

A. Dataset

We downloaded a dataset containing 3,256 images from Kaggle [30]. These photos were shot in the bone marrow lab at the Taleqani Hospital in Tehran, Iran. There are categories for benign and malignant data in the dataset. The Early, Pre, and Pro ALL subtypes of malignant lymphoblasts comprise the malignant class, while hematogenous pictures belong to the benign class. This dataset contains 512 images in the innocuous class, 979 images in the early class, 796 images in the Pro class, and 955 images in the Pre class. Every picture was captured with a Zeiss camera set up atop a microscope with a 100x magnification, and it was saved as a JPG file. A specialist used flow cytometry equipment to accurately identify and classify the cell kinds and subtypes. The images that have been gathered successfully undergo preprocessing next. At this stage, the images are resized in MATLAB R2023a. Resizing is done to enable the image to fit the dimensions of the input model that was used. ResNet50 and Inception-V3 were the models employed in this investigation, resulting in an image resolution reduction to 224x224 pixels for ResNet50 [22] and 229x229 pixels for Inception-V3 [25]. We used 976 photos for model testing, 228 images for validation, and 2052 images for training in our study.

Following the resizing process, the image underwent left-to-right and up-down flips. The output of this augmentation is fed into the model that is employed. ResNet50 and Inception-V3, two models that can classify 1000 different object categories and have acquired training from more than a million pictures, are the models that are used. sdgm is the optimization algorithm used in this training process, with a 0.001 learning rate, and a mini batch size of one twenty eight. The fine-tuning concept, a transfer learning technique, was applied in this work. It entails swapping out the pretrained output layer for a layer that contains the number of classes of cancers of the prostate cell datasets. The fully connected layer and the classification output layer are the layers that are changed.

B. Resnet 50

Microsoft Research unveiled ResNet-50, a convolutional neural network (CNN) with residual learning blocks that allow for more deeper network training. ResNet-50 showed exceptional performance [22], setting new benchmarks for accuracy and scalability at the 2015 ImageNet Large Scale Visual Recognition Challenge. With its 50 layers [22] and mostly residual block architecture, ResNet-50 is well-suited for a variety of computer vision applications, including segmentation, object identification, and picture classification. The ResNet-50 model, which provides excellent performance and resilience, has emerged as a mainstay in deep learning research and applications [22].

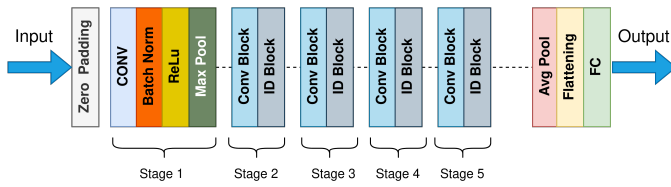


Fig. 1. ResNet50 model Architecture

As depicted in Figure 3, ResNet-50 comprises several residual blocks [25], each containing convolutional layers followed by identity mappings or projection shortcuts [?]. These blocks facilitate the flow of information across layers [25], alleviating the vanishing gradient problem and enabling efficient training of deep networks. With its skip connections and global average pooling layer, ResNet-50, achieves state-of-the-art results in image recognition tasks [25]. Additionally, its modular architecture and pre-trained weights make it a popular choice for transfer learning and rapid prototyping in various domains.

In ResNet-50's training process, (ReLU) activations are applied after each convolutional layer [25], preserving non-linearity and aiding in feature extraction. Furthermore, batch normalization and dropout techniques are utilized to enhance model generalization and prevent overfitting [25], contributing to its robustness and adaptability.

ResNet-50 offers significant advantages in deep learning applications [22], providing a solid foundation for building high-performance computer vision systems. By leveraging its hierarchical feature representations and fine-tuning capabilities [25], practitioners can expedite model development and achieve impressive results across diverse datasets and tasks.

C. Inception-v3

Google researchers unveiled Inception-v3, which is a major improvement over the original Inception architecture [27]. Inception-v3 builds on the fundamental ideas of its predecessor and includes a number of significant improvements meant to boost productivity and performance in picture categorization jobs. The improved Inception module, which underwent improvements to better optimise the network's capacity to extract and integrate characteristics from input pictures, is at the heart of Inception-v3. Convolution factorization, which divides huge convolutions into smaller ones to preserve expressive power while reducing computing complexity, is a notable addition to Inceptionv3 [27]. This factorization technique makes training with fewer parameters easier while simultaneously increasing computing efficiency. Furthermore, batch normalisation is still used by Inception-v3 [28]. Batch normalisation reduces internal covariate shift by normalising activations inside each mini-batch, which enhances generalisation performance. The inclusion of auxiliary classifiers at intermediate layers further contributes to the network's ability to learn robust features and combat overfitting [25]. These supplemental classifiers offer more

supervision signals during training, encouraging the network to learn discriminative features while serving as regularization mechanisms.

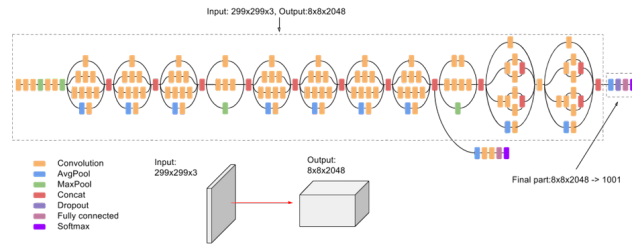


Fig. 2. Inception V3 model architecture

Incorporating reduction blocks, Inception-v3 strategically reduces the spatial dimensions of feature maps while increasing the number of channels [27]. This decrease in processing expense is made possible by a mix of convolutional

and pooling layers, preserving important information while downsampling feature maps. Moreover, the module of Inception

in Inception-v3 undergoes enhancements such as the introduction of bottleneck layers, which consist of 1x1 convolutions followed by larger convolutions [25]. This architectural design helps further reduce computational cost while maintaining representational capacity.

In addition to architectural improvements, Inception-v3 benefits from advanced training techniques, including data augmentation, dropout regularization, and gradient clipping [29]. These techniques contribute to better generalization performance and improved robustness against overfitting, ultimately enhancing the network's ability to classify images accurately across diverse datasets. Overall, Inception-v3 represents a significant milestone in deep learning architectures for image classification, demonstrating the continual advancement and refinement of convolutional neural networks for real-world applications.

D. Analysis

At this point, training and testing are the two phases of analysis. In order to determine which model is best, an analysis is conducted to evaluate and compare the models that have been used. A comparison of speed, accuracy of the model in categorising images into multiple preprocessed classes is the purpose of the assessment. Only the accuracy outcomes and the model's training time are used to compare models during the process of training. This evaluation determines which model can classify images with the highest accuracy. Additionally, a confusion matrix is used during the testing process to more precisely assess and contrast the models' performance. Accuracy, precision, sensitivity and f1-score are the four performance measurements. —make up the assessment parameters in testing.

III. RESULTS AND DISCUSSION

We have evaluated the model's output using multiple performance metrics. A machine learning algorithm's performance can be evaluated using performance metrics, which are essential measurements [24]. Performance metrics are a useful tool for evaluating multiple models' performances on a given data set and determining which model produces the best results. Metrics of performance may be utilised to assess the overall accuracy of the model and identify areas for improvement. When it comes to deep learning and transfer learning strategies, the most popular performance metrics are

- 1) accuracy
- 2) precision
- 3) sensitivity
- 4) f1-score

The dataset divided into four separate folders, each representing a different stage of acute lymphoblastic leukemia (ALL), was used to thoroughly train our model. This method gives our model the ability to identify ALL in any stage of the disease, allowing it to evaluate and forecast images from any point in the condition's progression. Furthermore, we have carried out an extensive comparative evaluation of these two pre-trained models, Resnet50 and Inceptionv3, to evaluate their efficacy with respect to ALL detection. This extensive assessment offers insightful information about how well these pretrained neural networks work to improve the predictive accuracy and general robustness of our model for this crucial medical use. The Inception-V3 and Resnet50 models training data results are displayed in the graphs below. Both the models

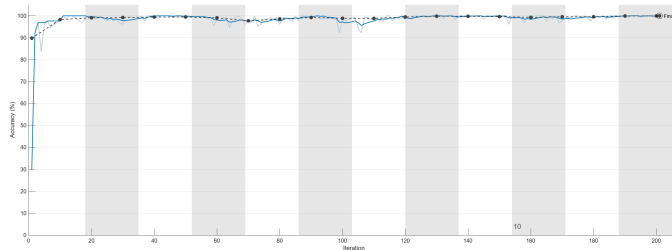


Fig. 3. ResNet50 training progress

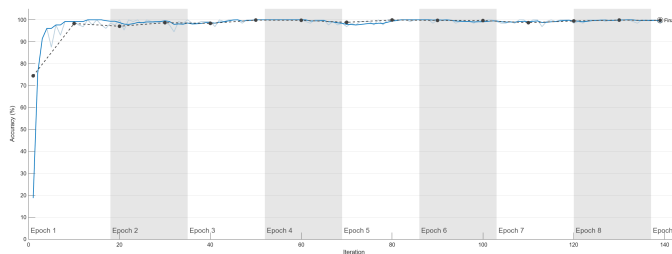


Fig. 4. Inception-v3 training progress

achieved 99.94 % and 99.82% accuracy respectively, indicates that both models have learned the training data very well. These high accuracies suggest that the models have effectively captured the patterns and relationships present in the

training dataset, leading to accurate predictions on the training examples. Following the training analysis, measurements and evaluations were also conducted on the testing that is shown in Table I. There are four parameters used, namely accuracy, sensitivity, precision and f-score.

TABLE I
PERFORMANCE METRICS OF RESNET50 AND INCEPTION V3

Metric	ResNet50 (%)	Inception v3 (%)
Accuracy	100	99.69
Precision	100	99.33
Sensitivity	100	98.63
F1 Score	100	98.96

A Confusion matrix is used in this evaluation to compare and select the best model. ResNet50 and Inceptionv3 confusion matrices are shown below. When compared to Inceptionv3,

Confusion Matrix

Benign	143 15.3%	2 0.2%	0 0.0%	0 0.0%	98.7%
Early	1 0.1%	294 30.1%	0 0.0%	0 0.0%	99.7%
Pro	0 0.0%	0 0.0%	289 29.9%	0 0.0%	100%
Pro	0 0.0%	0 0.0%	0 0.0%	241 24.7%	100%
	99.3%	99.3%	100%	100%	99.3%

Target Class

Fig. 5. Confusion Matrix for ResNet50

Confusion Matrix

Benign	143 15.3%	2 0.2%	0 0.0%	0 0.0%	98.7%
Early	1 0.1%	294 30.1%	0 0.0%	0 0.0%	99.7%
Pro	0 0.0%	0 0.0%	289 29.9%	0 0.0%	100%
Pro	0 0.0%	0 0.0%	0 0.0%	241 24.7%	100%
	99.3%	99.3%	100%	100%	99.3%

Target Class

Fig. 6. Confusion Matrix for Inceptionv3

the transfer learning model ResNet50 showed greater precision and accuracy ratings. By using the information gleaned from a sizable dataset, the ResNet50 model is able to identify blood cells more precisely and with more accuracy.

IV. CONCLUSIONS

This research investigated the performance of ResNet50 and Inception-V3 pre-trained models for the classification of leukemia cells. Both models demonstrated exceptional accuracy in both training and testing phases, achieving rates of 99.9% and 99.82%, respectively. The confusion matrix analysis further confirmed the effectiveness of both models, with

ResNet50 exhibiting slightly superior performance, achieving 100% accuracy, sensitivity, F1 score, and precision. However, Inception-V3 required a longer training time compared to ResNet50.

A. Future Scope

Ensemble Methods: Experiment with ensemble methods, such as stacking or bagging, to combine the strengths of multiple models and potentially improve performance. **Transfer Learning:** Investigate the effectiveness of transfer learning with other pre-trained models or fine-tuning existing models on larger leukemia datasets. **Explainable AI:** Develop techniques to explain the decision-making process of the models, providing insights into the features that contribute to the classification. **Real-time Applications:** Explore the feasibility of deploying these models in real-time clinical settings for rapid and accurate leukemia diagnosis. **Hybrid Approaches:** Combine deep learning with traditional machine learning techniques for complementary strengths. **Multi-modal Analysis:** Incorporate additional modalities, such as genetic data or clinical information, to enhance classification accuracy and provide a more comprehensive understanding of leukemia.

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